Detecting State-backed Information Operations on Twitter

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1. Introduction

The rise of Social Media has triggered the rise of state-backed information campaigns. These campaigns are often ran by specialized agencies, involving multiple accounts and years of planning with the goal to influence social media. This phenomenon has become relevant in recent years due to the Russian Internet Research Agency's (IRA) involvement with the 2016 U.S. election. Through the use of propaganda state-backed information campaigns seek to influence users on social media platforms towards their perspective. Identifying these state-backed campaigns is an ongoing and important challenge for these platforms.

Twitter's Election Integrity office serves to "strengthen Twitter against attempted manipulation, including malicious automated accounts and spam, as well as other activities that violate our Terms of Service."[1] Recently Twitter testified in front of congress about their efforts to stop election interference and if their platform's operations were influenced by politics.[2] To address these efforts twitter has begun disabling accounts that could potentially be associated with state-back information campaigns. They recently released two large datasets of all the accounts they disabled under the suspicion of foreign influence.[3] My analysis will focus on the dataset of deleted accounts and tweets suspected to belong to the IRA. The IRA is a private Russian company known for their online influence campaigns in elections world wide.[4]

While various analysis has been conducted on tweets to determine sentiment and bot status, my research endeavours to classify tweets belonging to the IRA. This can more broadly be extrapolated to detecting tweets involving propaganda. I developed 3 models using recurrent neural networks (RNN) to classify tweets belonging to the IRA. While the results were promising, and a significant improvement over the baseline, more fine tuning must be done to ensure a greater recall of propaganda tweets.

2. Methods

The dataset consisted of 28,960 tweets randomly sampled from two different sources. 6545 (22.6%) belong to the twitter IRA dataset while 22,415 (77.4%) control tweets were obtained from ongoing research conducted by the University of Michigan, Department of Sociology. The control tweets were randomly sampled from 1 billion

[1] Twitter. "Elections Integrity"
[2] Shepardson, David, Zengerle, Patricia. "Facebook, Twitter face U.S. Congress over politics and the internet"
[3] Twitter. "Enabling further research of information operations on Twitter".
[4] Lapowsky, Issie. "Facebook may have more Russian troll farms to worry about"

tweets belonging to the timeline of any user who has used #MeToo over the past 6 years. The IRA dataset includes all public, non-deleted tweets from accounts that twitter believed were connected to state-backed information campaigns. Tweets deleted by IRA users were not included in the dataset, but they represent less than 1% of overall activity. The tweets included a variety of different languages, predominately Russian and English.

To prepare the dataset for modeling, I applied natural language processing (NLP) techniques to the tweet text. This involved removing stop words, punctuation, emoji's numbers and any other special characters. Only English stop words were removed from the model. The text was then tokenized, producing a word frequency count. Timestamps were also included in the dataset, which were converted into a standardized datetime format as they came in two different formats. Once the dataset was cleaned I split it into a training and test set, reserving 25% for testing.

The methodological motivation was two-fold firstly to expand the current methods used in propaganda detection which little has been done with deep learning and to test existing well known approaches in text classification.[5] I built 3 different RNN models and evaluated their effectiveness on the assembled dataset.

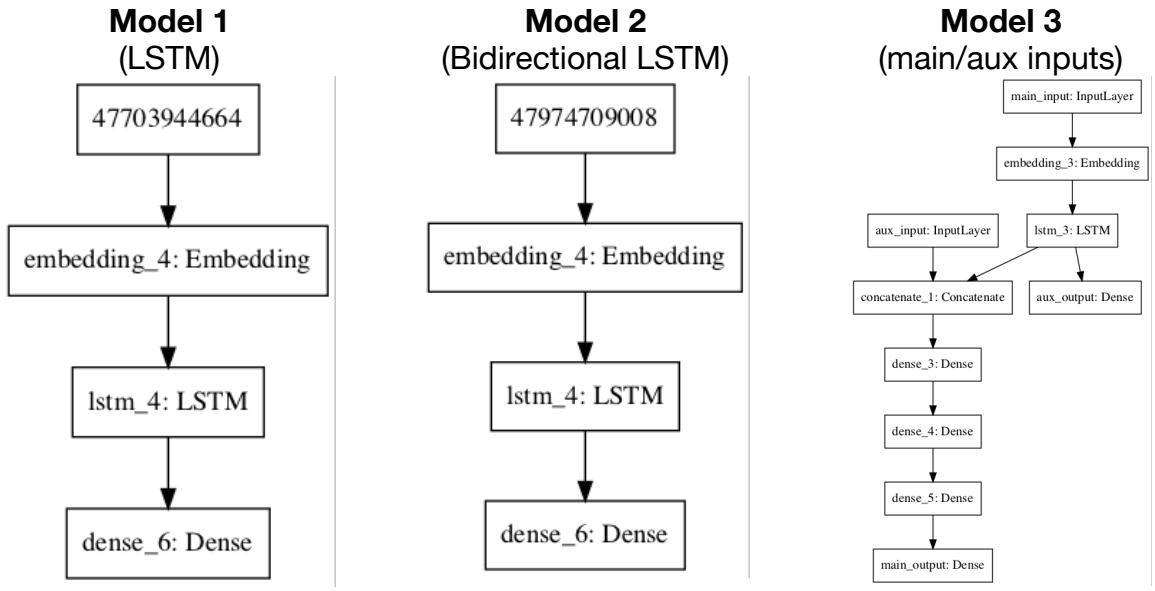


Fig. 1 Overview of Models

The first step in all models was to compute the word embedding's for the dataset. An embedding layer is a dictionary that provides a mapping of integer indices to dense

[5] Im, Jane. "We think the Russian Trolls are still out there: attempts to identify them with ML"; Ajao, Oluwaseun, Deepayan Bhowmik, and Shahrzad Zargari. "Fake news identification on twitter with hybrid cnn and rnn models."

vectors.[6] This presented two options: (1) learn the word embedding's jointly with the classification task in the model or (2) use a pre-trained embedding and import that into the model. I opted for option (1), allowing the model to jointly learn the weights. A pre-trained model would not be able to handle the new terminology and hashtags introduced within propaganda, eliminating key features. All embedding layers ere calculated with an input dimension of 10,000 words, an output dimension of 150 and an input length at 150 words.

Once the embedding layer was computed, next was to select a RNN layer. RNN are the best choice when dealing with sequential data due to their internal memory allowing for a deep understand of the sequence and its context. Choosing between a SimpleRNN and a LSTM (Long Short-Term Memory), I selected a LSTM because of the well known accuracy improvement provided over SimpleRNNs. (Which is mostly attributed to it's carry track, allowing for past sequence information to be introduced at later points while training.) The output dimensionality of all LSTM layers was limited to 32 units to avoid over fitting. One model also tested the use of a Bidirectional RNN which takes in the LSTM layer and replicates it for a reserved input. The motivation for this layer was to attempt to capture greater representations of the data, other than chronological.

While model 1 and 2 were variations on a basic LSTM, model 3 incorporated datetime stamps for each tweet, concatenating the output of the LSTM and the inputted datetime dimension. Then passing those representations through several dense layers, ultimately producing two outputs: main and aux, which will be explored in the evaluation section.

Code Summary

1. Given a set of raw tweets from two datasets, I merged the tweets and used NLP to clean them. Due to the size of the datasets, I sampled the tweets to get an approximately ¼ positive ratio.
2. Incorporating the two datasets I generated a test/train split of the sampled data for analysis. This split was done to be able to run *model.evaluate*, testing the performance of each model on an unseen set.
3. I created 3 different RNN models as well as a baseline, evaluating each on the test set. For each model I created a confusion matrix, classification report and an ROC curve.

3. Evaluation & Analysis

The study compared the performance of three models against each other and a baseline to investigate which produced a higher accuracy when on the test set. The baseline was calculated by selecting the most common class, which was the control

[6] Collins-Thompson, Kevyn. SI 670 Lecture Slides. "Part 2: Visualizing ConvNets, RNNs and LSTMs for sequence learning"

group. The baseline model had an accuracy of 77.65%. (Fig. 2) While test set accuracy was important metric, I also investigated when possible the precision, recall and f1-score. I was particularly interested in the class level recall values, as this represents the models ability to find all positive samples. Since model 3 was functional, only prediction values were presented, rather than binary class predictions, so it was excluded from the classification report.

Through analysis it is evident that model 3 has the same performance as the baseline, only predicting the dominant class. This is further supported in the confusion matrix. (Fig. 4) Future analysis must be done to understand why the main output of model 3 has an accuracy of 77.65%, but the auxiliary output an accuracy of 87.55%. Model 3 incorporated the datatime stamp of each tweet to add another layer of information for the neural net. Unfortunately, more debugging must be done to isolate where the error is occurring. More data should typically result in a higher test set accuracy, rather than the baseline.

Model	Test Accuracy	Scalar Test Loss
Baseline	77.65%	N/A
Model 1 (LSTM)	90.01%	.2926
Model 2 (Bi-LSTM)	89.2%	.3040
Model 3 (Main/Aux)	77.65%	3.60

Fig. 2 Model Accuracy/Loss Summary

On top of evaluating test accuracy, recall was an especially important metric. Recall represents the classifiers ability to find all positive samples. I analyzed the recall specifically for IRA tweets on model 1 and 2, the recall score was .67 and .68 respectively. This indicates that while the models perform well at identifying the control group tweets it erroneously misclassifies the IRA tweets. This is evident in the confusion matrices that approximately ~31% of the data is predicted to be within the control group, when in reality, it was an IRA tweet. (Fig. 4) This is most likely due to over fitting of the data.

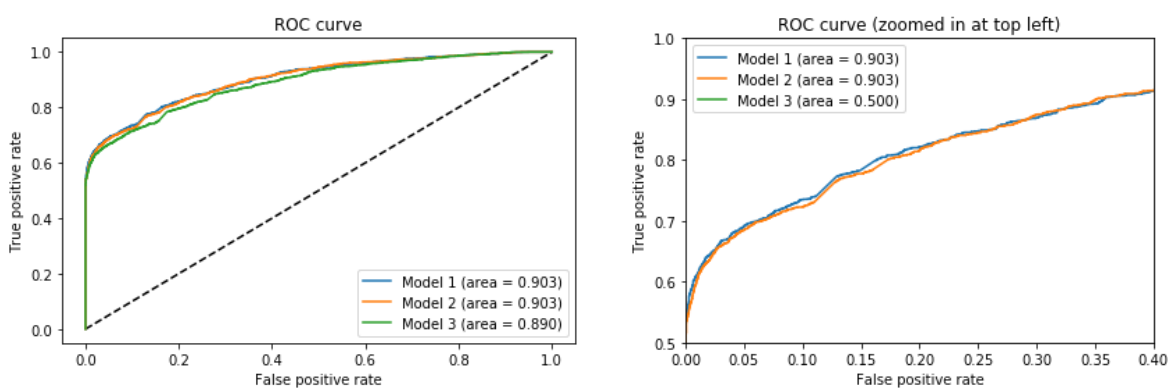
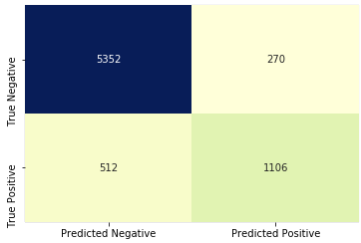


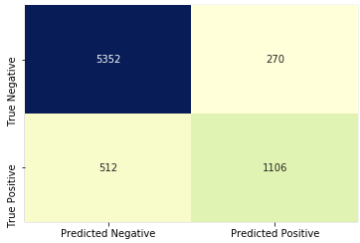
Fig. 3 ROC Curve for Models 1 & 2

Evaluating the performance of model 1 and 2 further, we can see that there is only a very small functional different between the two. The shows that in this situation adding the bidirectional layer to the LSTM has little performance boost. The ROC curve has been plotted for all models, it shows that model 1 and 2 have very similar true/false positive rates. (Fig. 3) The ideal curve should seek to have a false positive rate close to zero and a true positive rate close to one. We can see that when the True positive rate begins to approach .60 the false positive rate increases significantly.

Model 1
(LSTM)



Model 2
(Bidirectional LSTM)



Model 3
(main/aux inputs)

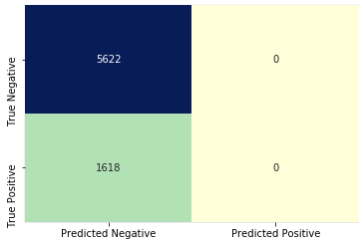
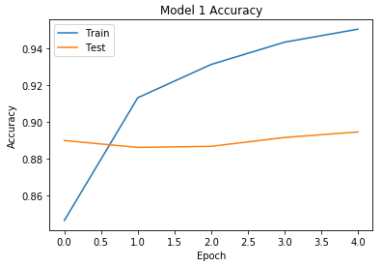


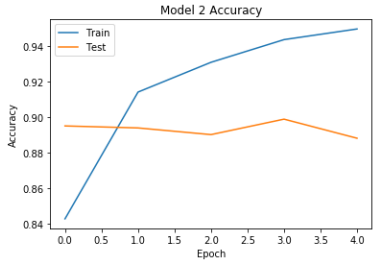
Fig. 4 Confusion Matrix

All models were trained on 5 epochs, which is an arbitrary cut off used for training and validation of the model. Each model went through 5 random passes of the data with a 70% random validation split. When plotting the model loss for the train/test split at each epoch, it's clear that model 1 and 2 are over fitting on the training set, showing little to no improvement of accuracy as the model progresses. (Fig. 5) The scalar loss of the test set on model 2 begins to increase when the training loss minimizes close to .15, showing a strong indication of over fitting. Over fitting could be caused by a number of factors, currently the output dimension of the LSTM in all models is 32, tweaking that could have an impact or potentially changing the input length of the tweets. Currently the input length is set to 100, which may be too high and producing strange outliers as the majority of tweets do not include that many words.

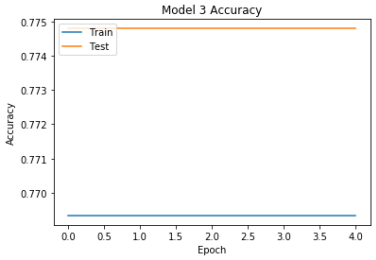
Model 1
(LSTM)



Model 2
(Bidirectional LSTM)



Model 3
(main/aux inputs)



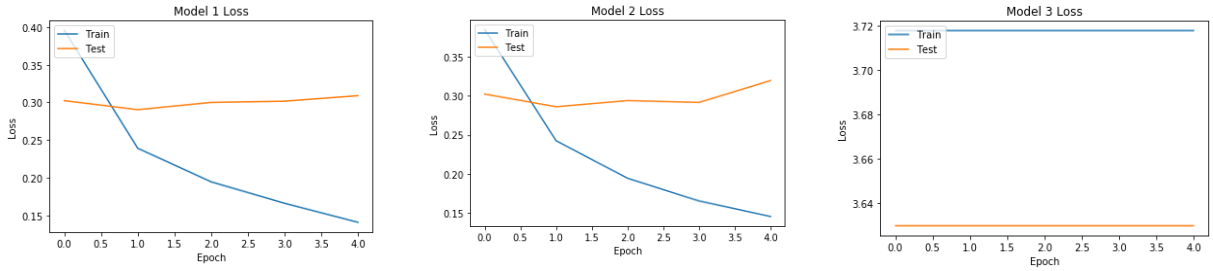


Fig. 5 Accuracy & Loss Training Plots

Overall model 1 performs the best across all evaluated approaches with a ~90% on the test set and a maximum of ~94% on the testing validation set during training. Due to potential input errors, model 3 does not function as intended. Model 1 could be further improved by reducing the dimension space, this can be done by: (1) changing the maximum number of words to tokenize in each string, (2) looking at less than 10,000 words for the embedding matrix or (3) outputting less than 150 dimensions from the embedding layer. All are different mechanisms to tweak the input/output which will impact the problem space.

4. Related Work

Studying twitter data is not a new phenomenon, primarily the goal was been to identify malicious automated bots. VS Subrahmanian et. al. explored this in a DARPA competition involving multiple authors and methods to detect twitter bots. They used classification, unsupervised outlier detection and network analysis, working with the features they engineered off of the base data. Most approach's scored well but required lots of manual feature engineering.[7] New studies have come out detailing the use of deep learning to study twitter data. Ajao et. al. tested three different deep learning models to detect fake news on twitter, with a peak accuracy of 82.29%.[8] Their finding was similar to mine such that the basic LSTM out performed the other variations. Kudugunta et. al. also applied deep learning to twitter data to detect bots, but engineering their own features and created an embedding outside of the neural net. Their tweet classifier had an overall accuracy of ~95% when they fed their own embedding's into the LSTM mode.[9] Only recently have models begun to analyze tweet content, independent of other features.

5. Discussion & Conclusion

State-backed information campaigns are a pressing, cross-platform issue that will only increase prominence. New techniques must be explored to detect and disable

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[7] Subrahmanian, V. S., Amos Azaria, Skylar Durst, Vadim Kagan, Aram Galstyan, Kristina Lerman, Linhong Zhu et al. "The DARPA Twitter bot challenge."

[8] Ajao, Oluwaseun, Deepayan Bhowmik, and Shahrzad Zargari. "Fake news identification on twitter with hybrid cnn and rnn models."

[9] Kudugunta, Sneha, and Emilio Ferrara. "Deep Neural Networks for Bot Detection." arXiv preprint arXiv:1802.04289 (2018).

accounts with malicious intent. This study sought to apply deep learning methods to determine if the content of a tweet alone was enough information to classify propaganda by the IRA. While an accuracy of 90% is promising, the low recall score shared across all models shows that future work must be done to expand on this approach. This could involve potentially adding in new variables such as IP address, follower count, tweet popularity and user level features. I believe that through the incorporation of more features, the accuracy and recall would increase.

All improvements aside, this study has shown that a relatively simple embedded RNN can detect to some degree the dialect of the Russian IRA. Further analysis of the embedding layer could yield unique word associations or heuristics that shed light on the relationship between language and tweet classification. While this area has been explored extensively within sentiment analysis, this field of classifying specific user group patterns is slowly emerging. As Ajao et. al. explored in their recent study, various various LSTM approaches were able to detect fake news without much required domain knowledge.[10]

This project provided me with functional knowledge of how to implement and evaluate neural nets with Keras. Over the course of the past month I have worked to establish a baseline model, test various iterations and implement a complex multi-input model. This has taught me the intricacies of RNN, specifically the trade offs between a SimpleRNN and LSTM. Future work should be conducted on this topic to explore the development of different variations of the LSTM models and the modification of different inputs/outputs to control for dimensionality. Given more time I would endeavor to improve the performance of model 3, incorporating more features to derive a better representation. I would also explore if ~90% is the actual performance cap on tweet text alone, or through further tweaking it could be improved without over fitting.

[10] Ajao, Oluwaseun, Deepayan Bhowmik, and Shahrzad Zargari. "Fake news identification on twitter with hybrid cnn and rnn models."

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